

1

Problem

- Images are spatially redundant, so low mask ratios make reconstruction too easy and waste compute on visible content.
- Vision transformers need a self-supervised objective that scales cleanly to very large models.

4

Efficiency & scaling

- Encoding only one quarter of patches reduces the dominant quadratic attention workload and speeds training by at least 3x.
- The same recipe scales from ViT-Base to ViT-Huge without contrastive pairs, negatives or a teacher network.

2

Asymmetric design

ENCODE VISIBLE PATCHES ONLY

- Randomly remove about 75% of image patches; the heavy ViT encoder receives only the remaining 25% and no mask tokens.
- A small decoder inserts learned mask tokens and reconstructs all patches; the decoder is discarded after pretraining.

5

Evidence

- Fine-tuned ViT-Huge reaches 87.8% ImageNet-1K top-1, then the best result among methods using only that dataset.
- Pretraining transfers strongly to COCO detection and ADE20K segmentation and exceeds supervised pretraining.

3

Objective

RECONSTRUCT MISSING PIXELS

- The target is normalized pixel values for masked patches, optimized with mean squared error.
- High masking forces holistic understanding rather than copying nearby texture while preserving a simple learning signal.

6

Meaning & limits

SCALABLE VISUAL PRETRAINING

- MAE showed that sparse input and a high mask ratio turn plain reconstruction into an effective representation learner.
- Pixel recovery is not explicitly semantic; large-scale pretraining still costs substantial compute and targets static images.

MAE

Masked Autoencoders

He et al. | arXiv:2111.06377v3

Mask most image patches, encode only the visible minority, and reconstruct missing pixels with a lightweight decoder for scalable self-supervised ViT pretraining.

masked modeling

asymmetric encoder

self-supervision